FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

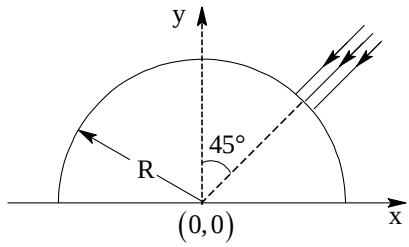
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. If a narrow beam of light falls on a glass hemisphere of refraction index 2 as shown in the figure. Then the image formed.



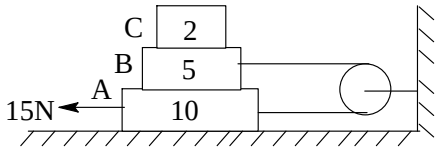
(A) Due to first surface has co-ordinates $\left(-\sqrt{2}R, -\sqrt{2}R\right)$

(B) Due to first surface has co-ordinates $\left(-\sqrt{3}R, -\sqrt{3}R\right)$

(C) Due to second surface has co-ordinates $\left(-R/\sqrt{2},\ R/\sqrt{2}\right)$

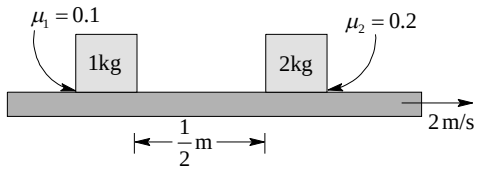
(D) Due to second surface has co-ordinates $\left(-R/\sqrt{3},\ R/\sqrt{3}\right)$

2. All surfaces are smooth, string is light and pulley is frictionless. Mark the correct statement(s):



(A) Acceleration of block A is 1 m/s^2
(B) Acceleration of block A and B will be same
(C) Acceleration of block C is zero
(D) Tension is string between pulley and wall is 10N

3. Two blocks kept on a rough plank are moving with the plank with velocity of $2\,\text{m/s}$. The plank is retarded uniformly for $1\,\text{sec}$ and brought to rest:
Assuming the plank to be sufficiently long then:



(A) The velocity of block of mass 2kg after collision with 1kg block is 1m/s

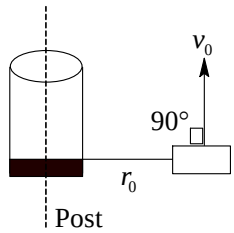
(B) The velocity of block of mass 2kg after collision with 1kg block is $\frac{1}{2}\text{m/s}$

(C) The final separation between two blocks is $\frac{1}{16}\text{m}$

(D) The final separation between two blocks is $\frac{1}{8}\text{m}$

Space for Rough work

4. A puck on a frictionless plane is given an initial speed v_0. The puck is attached to a string that wraps around a fixed vertical post:



(A) Mechanical energy of puck remains constant in the interval of motion
(B) Angular momentum of the puck about axis of post remains constant
(C) No torque acts on the puck about the axis of post.
(D) Work done by tension is zero.

5. A particle is executing SHM on a straight line. A and B are two points at which its velocity is zero. It passes through a certain point P (AP < BP) at successive intervals of 0.5 second and 1.5 second with a speed of 3 m/s:

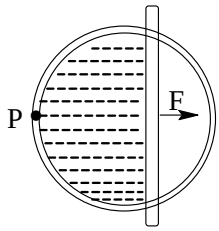
(A) The maximum speed of particle is $3\sqrt{2}\ m/s$

(B) The maximum speed of particle is $\sqrt{2}\ m/s$

(C) The ratio $\frac{AP}{BP}$ is $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

(D) The ratio $\frac{AP}{BP}$ is $\frac{1}{\sqrt{2}}$

6. A circular wire, 10cm in diameter, with a slider wire on it, is in a horizontal plane. A liquid film is formed, bounded by the wires, on the left side of the slider, as shown. The surface tension of the liquid is 0.1 N/m. An applied force $16\,\mathrm{mN}$, perpendicular to the slider, maintains the film in equilibrium. Ignore the sag in the film. What can be the distance between point F and slider?



(A) 8 cm
(B) 2 cm
(C) 5 cm
(D) Slider cannot be in equilibrium

Space for Rough work

7. 50 gm ice at $-10°C$ is mixed with 20 gm steam at $100°C$. When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 gm then: [Assume calorimeter was initially at 0°C, Take latent heat of vaporization of water = 540 cal/gm, Latent heat of fusion of water = 80 cal/gm, Specific heat capacity of water = 1 cal/gm-°C, Specific heat capacity of ice = 0.5 cal/gm-°C]

(A) Mass of water remaining is: $67.4\,\text{gm}$

(B) Mass of water remaining is $67.87\,\text{gm}$

(C) Mass of steam remaining is $2.6\,\text{gm}$

(D) Mass of steam remaining is $2.13\,\text{gm}$

8. 2 moles of He are mixed with 2 moles of H_2 in a closed adiabatic container. Initially the mixture occupies 3 litres at 27°C. The volume is suddenly decreased to $\left(\frac{3}{2}\right)$ liters. Choose the correct option(s):
(H_2 and He can be treated as ideal gases):

(A) γ for mixture is $\frac{3}{2}$

(B) Final temperature $= 300\sqrt{2}\,K$

(C) C_p for mixture is $2R$

(D) Work done in compression is totally converted into internal energy.

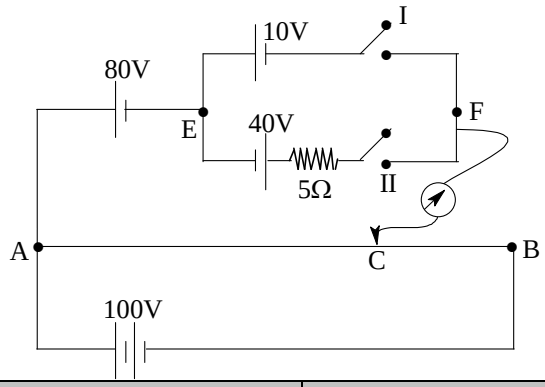
Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

In the potentiometer setup shown below all cells are ideal and all connecting wires offer zero resistance. Then answer the following question: (Take AB = 100 cm)



List – I		List – II	
(I)	Null deflection length AC	(P)	$90\,\text{cm}$
(II)	Current through 5Ω	(Q)	$40\,\text{cm}$
(III)	Current through 10V cell	(R)	$10\,\text{A}$
(IV)	Potential difference across EF	(S)	Zero
		(T)	$10\,\text{V}$
		(U)	$40\,\text{V}$
		(V)	$5\,\text{A}$

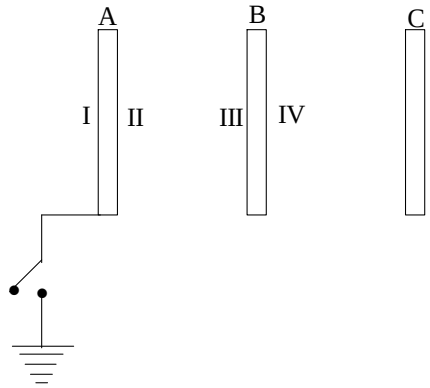
9. If only switch II is closed:
(A) $\text{I} \rightarrow Q,\ \text{II} \rightarrow \text{S},\ \text{III} \rightarrow \text{S},\ \text{IV} \rightarrow \text{U}$ (B) $\text{I} \rightarrow \text{P},\ \text{II} \rightarrow \text{S},\ \text{III} \rightarrow \text{U},\ \text{IV} \rightarrow \text{T}$
(C) $\text{I} \rightarrow \text{Q},\ \text{II} \rightarrow \text{S},\ \text{III} \rightarrow \text{S},\ \text{IV} \rightarrow \text{T}$ (D) $\text{I} \rightarrow P,\ \text{II} \rightarrow \text{S},\ \text{III} \rightarrow \text{S},\ \text{IV} \rightarrow \text{U}$

10. If both the switch are closed:
(A) $\text{I} \rightarrow \text{P},\ \text{II} \rightarrow \text{S},\ \text{III} \rightarrow \text{S},\ \text{IV} \rightarrow \text{S}$ (B) $\text{I} \rightarrow \text{Q},\ \text{II} \rightarrow \text{V},\ \text{III} \rightarrow \text{S},\ \text{IV} \rightarrow \text{S}$
(C) $\text{I} \rightarrow \text{P},\ \text{II} \rightarrow \text{R},\ \text{III} \rightarrow \text{R},\ \text{IV} \rightarrow \text{T}$ (D) $\text{I} \rightarrow \text{Q},\ \text{II} \rightarrow \text{R},\ \text{III} \rightarrow \text{R},\ \text{IV} \rightarrow \text{S}$

Space for Rough work

Answer the following by appropriately matching the list based on the information given in the paragraph.

Three rectangular conducting plates are placed closely parallel to each other as shown in figure. Plates A, C are given charges Q, and 2Q respectively.



The respective marking of surfaces is given on list I and List II provides with charges.

List – I		List – II	
(I)	I	(P)	$3Q/2$
(II)	II	(Q)	Zero
(III)	III	(R)	$-Q/2$
(IV)	IV	(S)	$Q/2$
		(T)	Q
		(U)	$-Q$

11. If switch is open:

(A) $I \rightarrow P,\ II \rightarrow R,\ III \rightarrow S,\ IV \rightarrow R$ (B) $I \rightarrow Q,\ II \rightarrow T,\ III \rightarrow U,\ IV \rightarrow U$

(C) $I \rightarrow S,\ II \rightarrow S,\ III \rightarrow R,\ IV \rightarrow S$ (D) $I \rightarrow P,\ II \rightarrow P,\ III \rightarrow R,\ IV \rightarrow S$

12. If switch is closed:

(A) $I \rightarrow P,\ II \rightarrow R,\ III \rightarrow S,\ IV \rightarrow R$ (B) $I \rightarrow Q,\ II \rightarrow T,\ III \rightarrow U,\ IV \rightarrow U$

(C) $I \rightarrow Q,\ II \rightarrow Q,\ III \rightarrow Q,\ IV \rightarrow Q$ (D) $I \rightarrow Q,\ II \rightarrow Q,\ III \rightarrow U,\ IV \rightarrow U$

Space for Rough work

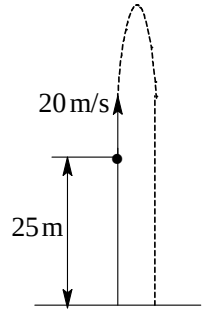
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

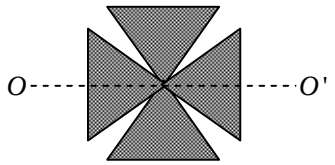
13. A box of mass 1kg is lying on a rough horizontal floor and a horizontal force acting on box has a magnitude of $F = 3t^2$N, where t is in seconds. If the box starts from rest, determine its speed (in m/s) at t = 3 second. The coefficient of static and kinetic friction between the box and horizontal floor are $\mu_s = 0.3$ and $\mu_k = 0.2$

14. Sunset is defined as the instant that the top of the sun dissappears below the horizon. How long is it from the time when the bottom of the sun hits the horizon until the instant of sunset assuming that you are standing on the equator on March 21? Call this time t_{sunset} and find it in seconds. Take diameter of Sun = 14×10^8 m and sun-earth distance = 1.5×10^{11} m. (take $\pi = 3.15$)

15. A boy throws a ball at time t = 0 with upward initial velocity = 20 m/s, from the corner of the roof of a building, whose height above the ground is 25 m. If the velocity of the ball is simply reversed after collision from the ground (magnitude remaining same), and the 3rd collision of the ball with the ground occurs at $t = t_1$ sec. What is the value of t_1?



16. Moment of inertia of an equilateral triangular plate of side length a, about an axis passing through one of its vertices perpendicular to its plane is equal to I_0. If the following symmetric structure is made from the same material, then its moment of inertia about the axis OO' is found to be equal to NI_0. Find N



17. A block is placed on a horizontal platform vibrating up and down, simple harmonically. It is observed that the block loses its contact with the platform when its angular frequency is 5 rad/s. The amplitude of oscillations cannot be less than 'A' cm, then find the value of A:

18. End A of a rod AB of length L = 0.5 m and of uniform cross-sectional area is maintained at some constant temperature. The heat conductivity of the rod is K = 11.34 $Wm^{-1}K^{-1}$. The other end B of this rod is radiating energy into vacuum and the wavelength with maximum energy density emitted from this end is $\lambda_0 = 75000\text{Å}$. If the emissivity of the end B, e = 1, determine the temperature of the end A (in Kelvin). Assume that lateral surface is insulated. (Take Wien's constant = 3×10^{-3} mK, Stefan's constant $\sigma = 5.67 \times 10^{-8}$ W/m^2K^4)

Space for Rough work

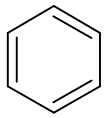
Chemistry

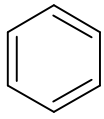
PART – II

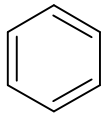
SECTION – A
(One or More than one correct type)

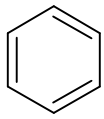
This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. If x is the number of chlorine atoms and y is the degree of unsaturation then which of these reactions will give a product with maximum value of $x + y$:

(A)  $+ \underset{(\text{excess})}{CH_3COCl} \xrightarrow[AlCl_3]{\text{anhyd}}$

(B)  $+ \underset{(\text{excess})}{Cl_2} \xrightarrow[\text{Cold, dark}]{\text{Anhyd. } AlCl_3}$

(C)  $+ \underset{(\text{excess})}{CH_3Cl} \xrightarrow[AlCl_3]{\text{Anhyd.}}$

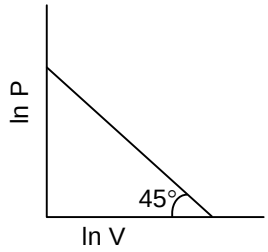
(D)  $+ \underset{(\text{excess})}{Cl_2} \xrightarrow[500\,K]{h\nu}$

20. Select the incorrect statement(s):
(A) Second period of modern periodic table contains maximum number of gaseous elements and sixth period contains maximum number of solid elements.
(B) $SrSO_4$ is the most stable towards heat among all the sulphates of alkaline earth metals.
(C) Molten hydrides of all alkali metals liberate H_2 gas at anode on electrolysis.
(D) The coordination number of Zn in Wurtzite structure is 4.

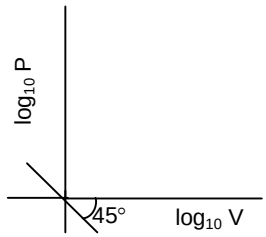
Space for Rough work

21. For one mole of an ideal gas at R Kelvin, which of the following graphs is/are correct (M. W. = 100) ($R = \frac{1}{0.0821}$ K) (P is in atm and V is in litre)

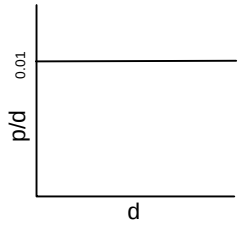
(A)



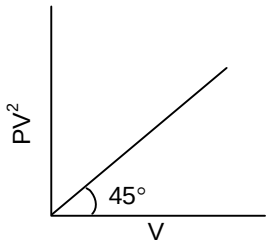
(B)



(C)



(D)



22. A solution is prepared by dissolving 1.5 g of a monoacidic base into 1.5 kg of water at 300 K, which showed a depression in freezing point by $0.165^{o}C$. When 0.496 g of the same base titrated, after dissolution, requires 40 ml of semimolar H_2SO_4 solution. If K_f of water is 1.86 K kg mol^{-1}, then select the correct statement(s) out of the following (assuming molarity = molality):
(A) The pH of the solution of weak base is 11.903
(B) Ionisation constant of the base is 8×10^{-4}
(C) The osmotic pressure of the aq. Solution of base is 2.167 atm
(D) The base is 10% ionized in aq. Solution

23. Complete the reaction:
$CS_2 + \text{Aq. NaOH} \rightarrow$ _____ + _____ + H_2O
(A) Na_2S (B) Na_2CS_3
(C) Na_2SO_3 (D) $Na_2\text{-}CO_3$

Space for Rough work

24. If for an equation like $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, the ΔG (reaction) expression given is

$$\Delta G = 4.73 \text{ KJ mol}^{-1} + 2.5 \text{ KJ mol}^{-1}\ \ell n\left\{\frac{P^2_{NO_2}}{P_{N_2O_4}}\right\} \text{ then}$$

(A) If small amount of N_2O_4 is introduced in a closed container, then reaction moves in forward direction spontaneously.

(B) Initially, ΔG° value is negative for the given decomposition.

(C) At equilibrium, $Q_P = K_P =$ antilog $\left[\frac{-\Delta G^0}{RT}\right]$

(D) If mole fraction of N_2O_4 >mole fraction of NO_2 still reaction can be spontaneous in backword direction.

25. Identify the correct statements from the following reactions:

(i) $SnCl_2 + H_2O \rightarrow X + HCl$

(ii)

$$SnCl_2 + 2NaOH \longrightarrow Y + 2NaCl$$
$$\downarrow 2NaOH(\text{excess})$$
$$Z + H_2O$$

(iii) $3SnCl_2 + 2AuCl_3 \rightarrow 3SnCl_4 + P$

(iv) $SnCl_2 + 2HCl + I_2 \rightarrow SnCl_4 + Q$

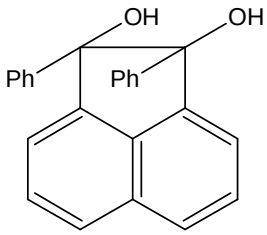
(A) X shows white turbidity and is basic

(B) Y is white ppt. and Z is soluble in water

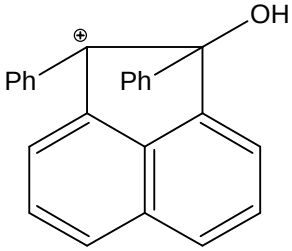
(C) P is a metal

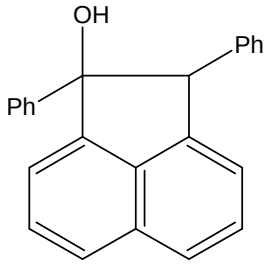
(D) Q is an acid

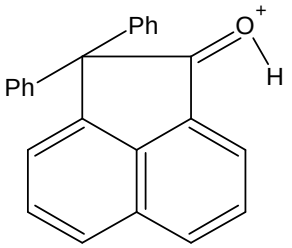
Space for Rough work

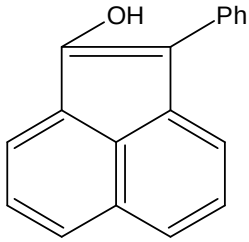
26. When 

is treated with H_2SO_4 and heated then what are the intermediate and products formed?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

Match the product of following reactions.

List – I		List – II	
(I)	$NH_2CONH_2 + HNO_2 \rightarrow$	(P)	N_2
(II)	$CaCN_2 + H_2O \rightarrow$	(Q)	CO_2
(III)	$CaOCl_2 + NH_3 \rightarrow$	(R)	H_2O
(IV)	$N_2H_4 + HNO_2 \rightarrow$	(S)	NH_3
		(T)	N_3H
		(U)	$CaCO_3$

27. Choose the correct options
(A) I-P,Q,S,T
(B) II-P,Q,U
(C) II-P,Q,S
(D) I-P,Q,R

28. Choose the correct options
(A) III-P,R,S,U
(B) IV-P,R,S
(C) IV-R,T
(D) III-P,R,T

Space for Rough work

Answer the following by appropriately matching the list based on the information given in the paragraph.

Match List – I with List – II.

List – I		List – II	
(I)	A blue solution	(P)	$Cu(OH)_2$
(II)	A blue precipitate	(Q)	$CuSO_4.5H_2O$
(III)	Red colour compound	(R)	$[Co(SCN)_4]^{2-}$
(IV)	Tetra coordinated compound	(S)	$KFe[Fe(CN)_6]$
		(T)	Pb_3O_4
		(U)	HgI_2

29. Choose the correct options
(A) III-P,Q,S (B) II-P,Q,S,T
(C) I-Q,R (D) IV-Q,R,T

30. Choose the correct options
(A) I-Q,R,S,U (B) II-Q,T,S,U
(C) III-T,U (D) IV-Q,R,U

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. An equilibrium $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ is obtained by starting with1 mole of PCl_5 in a 1L vessel. A vapour density of equilibrium mixture was found to be 250.2. Now, the equilibrium mixture is transferred into a vessel of 2L capacity, its new degree of dissociation is.

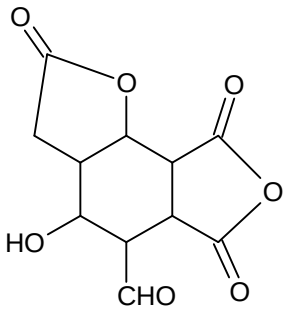
Space for Rough work

32. A glow-worm of mass 5.0g emits red-light (650 nm) with power of 0.10 W entirely in the backward direction. What velocity (v) in m/sec, will it have accelerated to after 10 years, if released in free space (and assumed to live). Then v is?

33. In the unbalanced reaction:
$Al + KMnO_4 + H_2SO_4 \longrightarrow KHSO_4 + Al_2(SO_4)_3 + MnSO_4 + H_2O$
If the stoichiometric coefficients of Al, H_2SO_4, $MnSO_4$ and H_2O are w, x, y and z respectively, the numerical value of $\left(\frac{x}{y}+\frac{z}{w}\right)$ will be

34. In acid rain in Delhi it was found that 1 litre rain water consist of 2.463 litre equimolar mixture of SO_3 and NO_2 gas dissolved at a partial pressure of 0.1 atm. Find out the pH of the solution if HNO_2 produced is 60% dissociated under given condition at $27°C$. (R = 0.0821 litre atm/mole K) (log 1.8 = 0.25)

35. If molecule shown below has molar mass→x then find the mass of CH_3MgBr (in grams) which can react (max.) with $\frac{x}{2}$ grams of compounds $[Mg\text{-}24.3, Br\text{-}80]$



36. $H_2C{=}C{=}O \xrightarrow{(1)\,Br_2\,(anddition)1eq.} A \xrightarrow[2eq.]{CH_3MgBr} B$ suppose both reactions occur by 100% efficiency then the molar mass of B:

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $S_1=\sum_{r=0}^{n}{}^nC_r\left(-\frac{1}{2}\right)^r$; $S_2=\sum_{r=0}^{n}{}^nC_r\left(-\frac{3}{4}\right)^r$ .....$S_m=\sum_{r=0}^{n}{}^nC_r\left(-\left(\frac{2^m-1}{2^m}\right)\right)^r$ then:

(A) $\sum_{k=1}^{4}S_k=\frac{2^{4n}-1}{2^{4n}\left(2^n-1\right)}$

(B) $\sum_{k=1}^{16}\left(S_k\right)^{\frac{1}{n}}=1-\frac{1}{2^{16}}$

(C) $\sum_{k=1}^{4}S_k=1-\frac{1}{2^{16}}$

(D) $\sum_{k=1}^{16}\left(S_k\right)^{\frac{1}{n}}=\frac{1}{15}\left(1-\frac{1}{2^{16}}\right)$

38. If a, b, x, y are positive natural numbers and $\frac{1}{x}+\frac{1}{y}=1$ then $\frac{a^x}{y}+\frac{b^y}{x}$ is same as:

(A) $\frac{xa^y+yb^x}{x+y}$

(B) $\frac{a^x+b^y}{x+y}$

(C) $\frac{a^x}{x}+\frac{b^y}{y}$

(D) $\frac{a^y}{x}+\frac{b^x}{y}$

39. If $\lim_{n\to\infty}\left(\frac{1}{1^2}+\frac{1}{2^2}+.....+\frac{1}{n^2}\right)=\frac{\pi^2}{6}$ *and* $S=\lim_{n\to\infty}\left(\frac{1}{1^3.2^3}+\frac{1}{2^3.3^3}+\frac{1}{3^3.4^3}+.....+\frac{1}{n^3.\left(n+1\right)^3}\right)$,

then $S+\pi^2$ is equal to

(A) A non-integral number

(B) An rational number

(C) A transcendental number

(D) An integer

Space for Rough work

40. The ends of a line segment are $P(1,3)$ and $Q(1,1)$. R is a point on the line segment PQ such that $PR:QR=1:\lambda$. If R is an interior point of the parabola $y^2=4x$, then

(A) $\lambda \in (0,1)$ (B) $\lambda \in \left(-\frac{3}{5},1\right)$

(C) $\lambda \in \left(\frac{1}{2},\frac{3}{5}\right)$ (D) None of these

41. $f:N\to A$ and $g:N\to B$ are two onto functions such that $f(n)=\left[n\sec^2\theta\right]$ and $g(n)=\left[n\operatorname{cosec}^2\theta\right]$, for some θ. If $\sec^2\theta$ is irrational, then (where $[.]$ denotes the greatest integer function)

(A) $A\cap B=\phi$

(B) $A\cup B=N$

(C) f and g are bijective

(D) $A\cap B$ is non-empty finite set

42. A triangle with vertices $P(2,5), Q(5,2)$ *and* $R(-1,-1)$ is inscribed in the rectangular hyperbola $xy-x-y-3=0$ and let $H\equiv(\lambda,\lambda)$ is orthocentre of ΔPQR, let A, B *and* C be the feet of perpendicular from P, Q and QR, RP and PQ respectively and r is in radius of ΔABC, then

(A) Circumcentre of ΔPQR is $\left(-\frac{3}{2},-\frac{3}{2}\right)$

(B) Incentre of ΔABC is $(3,3)$

(C) Equation of AB is $x+y-\frac{26}{5}=0$

(D) $r=\frac{2\sqrt{2}}{5}$

Space for Rough work

43. The x that satisfying the equation $\left(2\sec x-1\right)^{\frac{1}{3}}+\left(\sec x-1\right)^{\frac{1}{3}}=1$ can be in the interval

(A) $x\in\left(-\frac{\pi}{3},\frac{\pi}{3}\right)$ (B) $x\in\left(-\frac{\pi}{6},\frac{\pi}{12}\right)$

(C) $x\in\left(\frac{3\pi}{4},\frac{5\pi}{4}\right)$ (D) None of these

44. Let all the letters of the word 'MATHEMATICS' are arranged in all possible order. Three events A, B and C are defined as:
A: Both M are together
B: Both T are together
C: Both A are together
Which of the following hold(s) good?

(A) $P\left(A\right)=P\left(B\right)=\frac{2}{11}$

(B) $P\left(A\cap B\right)=P\left(B\cap C\right)=P\left(C\cap A\right)=\frac{2}{55}$

(C) $P\left(A\cap B\cap C\right)=\frac{4}{495}$

(D) $P\left[\left(A\cap\overline{B}\right)/\overline{C}\right]=\frac{58}{405}$

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

Consider the following equations of curves:

$C_1 : x^2 + y^2 = 4$

$C_2 : y^2 = 8\sqrt{6}x$

$C_3 : \frac{x^2}{1} + \frac{y^2}{5} = 1$

$C_4 : \frac{x^2}{25} - \frac{y^2}{1} = 1$

List-I : Contains the equation of common tangent between two of the given curves
List-II: Contains some information

List – I		List – II	
(I)	$y = \sqrt{2}\left(x + \sqrt{6}\right)$	(P)	Is a chord of C_1
(II)	$y = \sqrt{3}x + 2\sqrt{2}$	(Q)	Is not a chord of C_4
(III)	$y = \frac{1}{\sqrt{3}}x + \frac{4}{\sqrt{3}}$	(R)	Common tangent of C_1 & C_3
(IV)	$y = x + 2\sqrt{6}$	(S)	Common tangent of C_2 & C_4
		(T)	Common tangent of C_1 & C_2
		(U)	Common tangent of C_2 & C_3

45. Which of the following is the only correct combination?

(A) $\text{II} \rightarrow \text{RQ}$ (B) $\text{I} \rightarrow \text{SP}$

(C) $\text{II} \rightarrow \text{PU}$ (D) $\text{I} \rightarrow \text{TP}$

Space for Rough work

46. Which of the following is the only correct combination?

(A) $\text{III} \rightarrow \text{SP}$ (B) $\text{III} \rightarrow \text{RQ}$

(C) $\text{IV} \rightarrow \text{TP}$ (D) $\text{IV} \rightarrow \text{SQ}$

Answer the following by appropriately matching the list based on the information given in the paragraph.

Consider the following functions:

$f(x) = \dfrac{32}{4 + x^2 + x^4}$

$g(x) = 9 + x^2$

List-I: Contains new functions defined with the help of given functions
List-II: Contains some information about them

List – I		List – II	
(I)	$p(x) = f(x) - g(x)$	(P)	Is an odd function
(II)	$q(x) = \left(f(x) + g(x)\right)x$	(Q)	Is an even function
(III)	$r(x) = f(x) \cdot g(x)$	(R)	Function never intersect X-axis
(IV)	$s(x) = \dfrac{f(x)}{g(x)}$	(S)	Range is a subset of $(0, \infty)$
		(T)	Range is a subset of $(-\infty, 0)$
		(U)	Function intersect X-axis exactly once

47. Which of the following is the only correct combination?

(A) $\text{II} \rightarrow \text{PST}$ (B) $\text{I} \rightarrow \text{QUS}$

(C) $\text{II} \rightarrow \text{PRU}$ (D) $\text{I} \rightarrow \text{QRT}$

48. Which of the following is the only correct combination?

(A) $\text{III} \rightarrow \text{QSU}$ (B) $\text{IV} \rightarrow \text{QS}$

(C) $\text{III} \rightarrow \text{QTR}$ (D) $\text{IV} \rightarrow \text{PS}$

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. The value of $\lim_{t\to 0} \ln\left(\frac{1}{t}\int_0^1 (1+2\sin 3x)^{4/x}\,dx\right)$ is equal to:

50. Consider a square matrix of order 2 given by $D_{r_1,r_2} = \begin{bmatrix} \tan^{-1}\frac{r_1}{r_2} - 4 \\ \cot^{-1}\frac{r_1}{r_2}\ 4 \end{bmatrix}$ if $\Delta_n = \sum_{r_1=1}^{n}\sum_{r_2=1}^{n} D_{r_1.r_2}$, then find the determinant value of the matrix A defined by $A = \lim_{n\to\infty} \frac{\Delta_n}{n^3 \sin\left(\frac{\sqrt{\pi}}{n}\right)}$

51. Let $y = f(x)\left(f: R \to R\right)$ be an explicit function defined by the implicit equation $x^3 + y^3 + 3\left(x^2 + y^2\right) + 3(x+y) = 14$ and g be the inverse of f. If $f\frac{d}{dx}\left(f\left(x + g(x)\right).g\left(x + f(x)\right)\right)$ at $x = -1 + \sqrt[3]{15}$ is equal $\lambda(15)^{2/3}$, where $\lambda \in I$, then find the value of $|\lambda|$:

Space for Rough work

52. If l_1 and l_2 are the least and greatest distance between parallel tangents drawn to the curve $f(x) = x - \sin x$ where $x \in \left(\frac{-\pi}{6}, \frac{13\pi}{6}\right)$, then find the value $(l_1 + l_2)$.

[Use: $\pi = 3.14$]

53. If $\sum_{n=0}^{\infty} \tan^{-1}\left(\frac{\cot^{-1}(n^2+3n+3)}{1+\cot^{-1}(n+1)\cot^{-1}(n+2)}\right) = \tan^{-1}\left(\frac{p}{4}\right)$ then find the value of $\cos^{-1}\left(\cos(p-1)\right)$.

[Use: $\pi = 3.14$]

54. If $z_1, z_2, z_3 \in \mathbb{C}$ satisfy the system of equations given by $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ *and* $z_1 z_2 z_3 = 1$ such that $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$ then find the value of $|z_1 + z_2^2 + z_3^3|$ is.

Space for Rough work